

Design of Combinational Digital Circuits Optimized with Ising Model and PSO Algorithm

Ying Li¹, Penglei Zhao¹, Bingrui Guo³, Chenhui Zhao¹, Xiaojie Liu^{1,2}, Shan He^{1,2}, Donghui Guo^{1,2*}

¹National Model Microelectronics College, Xiamen University, Xiamen, China

²IC Design R&D Engineering Center of Fujian Province, Xiamen, China

³Department of Mathematics, University of California, San Diego, America
23120191150224@stu.xmu.edu.cn; *dhguo@xmu.edu.cn

Abstract—Evolutionary circuit is an important part of electronic design automation. It is a kind of automatic circuit design system by intelligent algorithm, widely used in robot controller design, circuit design and other fields. Compared with other intelligent algorithms, Particle Swarm Optimization (PSO) has better performance in the process of circuit evolution, but it has the disadvantage of falling into local optimum easily during evolution, resulting in meaningless increase of computing resources. This paper proposes a hybrid algorithm based on Ising model and PSO algorithm for optimizing combinational logic circuits. The Ising model is a kind of stochastic process model describing the material phase transition and has the property that it can accept a worse solution than the current one with a certain probability, which can increase the diversity of particles and avoid particles trapped in local optima point during the evolutionary process. Experimental results show that the hybrid algorithm has better performance in terms of computational complexity and circuit area.

Keywords—combinational circuits; optimizational design; Ising model; PSO; hybrid algorithm

I. INTRODUCTION

Combinational logic circuit is a type of digital circuit whose output only depends on the current input and is independent of the state of the circuit. There is no feedback delay channel between the input and output of the combinational logic circuit, and there is no memory unit in the circuit. The truth table of a combined logic circuit is unique, but the corresponding logic circuit or logical expression may have many forms of implementation.

In actual work, in order to make the design of logic circuits more concise, designers have created many artificial design ways, such as Boolean algebra, Karnaugh Map (K_map)^[1], and Quine-McCluskey^[2]. Algebraic methods apply some known algebraic theorems to minimize logical functions. K_map is a graphical representation of a logical function in which variables can be eliminated by merging the smallest adjacent terms, but only a maximum of five variables can be minimized^[1]. McCluskey is a tabular method that can minimize any number of input variables^[2]. However, the execution time required for the McCluskey process has grown in an index with the number

of input signals. In addition, the Karnaugh map and McCluskey methods only use AND, OR, and NOT gates, and ignore all other gates.

The artificial design method greatly depends on the experience and ability of human designers, and is limited by their habits (conventions), imagination and creative thinking. With the development of technology, evolved hardware has gradually emerged. Evolving Hardware (EHW) is a method of automatically designing circuits based on natural phenomena^[3]. The design process of the evolutionary circuit is essentially different from the traditional design process, because it is not determined by the designer's knowledge and experience, it is determined by the evolution process. So far, many different methods of evolution have been created. For example, combinatorial circuits are designed using genetic algorithms^[4] and adaptive genetic algorithm^[5]. In addition, there are many papers using the PSO algorithm to design and optimize combinational logic circuits.

A large number of experimental data indicates that the PSO algorithm has better performance in the evolution of circuits. Coello first proposed the use of the PSO algorithm to design combinatorial logic circuits, proving that the PSO algorithm is better than human design and genetic algorithm^[6]. This algorithm is inspired by the research of bird flock foraging behavior. K.Sagar compared the performance of the PSO algorithm, genetic algorithm and ant colony optimization in designing combinatorial logic circuits^[7]. The experimental results show that the PSO algorithm can discover the feasible solution with the least gate number and the shortest time.

However, the PSO algorithm has its own limitations^[8], which makes it easy to fall into the local optimal point in the process of evolution logic circuit, instead of finding the global optimal point out of the loop, resulting in the meaningless increase of computing resources. Therefore, combined with the Ising model^[9], the algorithm in this paper can accept the solution worse than the current solution at a certain probability, so that the calculation can avoid falling into the local optimal point and search for the global optimal solution. In this paper, the PSO algorithm is used to guide the initial particle to change towards the optimal solution, and the energy function of the Ising model

is used to determine whether to accept this change, in order to avoid the initial particle in the evolutionary process to fall into local optimal points.

The remainder of this paper is organized as follows: In section II, we will introduce the preliminary work of the hybrid algorithm of Ising model and PSO. Section III introduces the detailed process of optimizing the combined circuit with the hybrid algorithm. In section IV, we will describe the experimental process and result analysis. Finally, the conclusion will be provided in section V.

II. PRELIMINARY WORK

The PSO algorithm first initializes a group of random particles, each of which has its position and speed. In each iteration, the particles update the personal position (p_best) it has reached, and the best position (g_best) in a set of particles, keep the particles close to the direction of p_best and g_best . The equation for the particle to update its velocity and position is as follows:

$$V_{P(N+1)} = W * V_{PN} + C_1 * rand_1 * (p_best - X_{PN}) + \quad (1)$$

$$C_2 * rand_2 * (g_best - X_{PN})$$

$$X_{P(N+1)} = V_{P(N+1)} + X_{PN} \quad (2)$$

where $V_{P(N+1)}$ and V_{PN} represent the current and past velocities of particle P respectively; $X_{P(N+1)}$ and X_{PN} represent the current and former positions of particle P; $rand_1$ and $rand_2$ are two random numbers; C_1 and C_2 are two acceleration constants; W is the inertial weight that controls the global and local exploration of the particle, and W decreases linearly with the increase of the number of iterations, thus ensuring that the particle can converge.

$$W = W_{min} + (W_{max} - W_{min}) * [(Iter_{max} - Iter) / Iter_{max}] \quad (3)$$

where W_{max} and W_{min} are the maximum and minimum values of weights; $Iter$ and $Iter_{max}$ represent the current iterations and the maximum iterations, respectively.

In order to enhance the diversity of particles during searching, we combine the Ising model to allow particles to accept a worse solution than the current solution in a certain probability. We set the energy function of the particles, defined as follows:

$$E_{(si)} = -J \sum_{\langle i,j \rangle} S_i S_j \quad (4)$$

where $E_{(si)}$ represents the energy value of the i th particle, J represents an energy coupling constant, S_i and S_j represent the output value of the truth table of the evolutionary circuit and the output value of the truth table of the target respectively. S_i and S_j value +1 means that under this input combination, the output of the truth table is 1; S_i and S_j value -1 means that under this input combination, the output of the truth table is 0. If the truth table of the evolutionary circuit and the target truth table correspond exactly and J is positive, the energy of the particle will be the minimum value within the value range.

The positions of particles after changes in (1) and (2) are denoted as $X_{P(N+1)}$. The energy values of each particle after

changes were calculated and compared with the energy values before changes.

where $P_{(si)}$ represents the probability of accepting a worse solution than the current one, and the calculation equation is defined as follows:

$$P_{(s_i)} = \exp\left(-\frac{\Delta E}{KT}\right) \quad (5)$$

The ΔE represents the energy difference between the particles before and after the change. K is a constant. T is the temperature value that adjusts the change of probability, which decreases with the increase of iteration number, indicating that with the increase of iteration number, the probability that accepts a worse solution is gradually decreasing, so that the algorithm can finally converge to an optimal solution. The equation for T is defined as follows:

$$T = T_0 * [(Iter_{max} - Iter) / Iter_{max}] \quad (6)$$

The initial value of T is set to $|E_{max} - E_{min}|$.

Because the Ising model can accept a worse solution than the current one with a certain probability to avoid the loss of diversity, we improve the PSO algorithm and propose a hybrid algorithm combining the Ising model and Particle Swarm Optimization (IPSO).

III. IPSO FOR DESIGNING THE LOGIC CIRCUITS

In this paper, the IPSO algorithm is used to design and optimize combinatorial logic circuits. The circuit connection form can be directly obtained according to a given truth table, and the algorithm can prevent particles to fall into a local optimal point during circuit evolution.

First, a group of particles are randomly initialized, and each particle corresponds to a circuit connection. The connection form of the combinational logic circuit is shown in Fig. 1.

Each unit in the mesh shown in Fig. 1 is a gate having two inputs and one output. The first column of the grid comes from the variable input, which is the input of the circuit. The input of all other columns comes from the output of the previous column. For example, X1, X2, X3 represent the output of the gate of the first column; Y1, Y2, and Y3 represent the output of the gate of the second column; Z1, Z2, and Z3 constitute a gate that generates the output of the entire circuit.

Encode a circuit as a matrix, each row of which represents a logical operation. The 3*3 grid represented by a 7*3 matrix. The first and third numbers in each row represent two inputs of this logic gate, and the second number represents the type of logic gate. Five logic gates here are used, such as AND, NOT, OR, WIRE and XOR. WIRE means the input equals the output. The program will traverse each row from top to bottom, corresponding to the column elements scanned from left to right in the grid, so the output of the first row of the matrix is X1 in Fig. 1. TABLE I and TABLE II describe the coding rules.

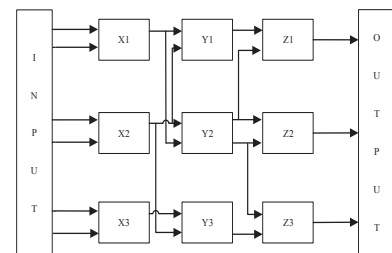


Figure 1. The connection form of the combinational circuit

TABLE I. MATRIX VALUE OF 2ND COLUMN

Matrix value of 2 nd column	Type of gate
1	AND
2	OR
3	XOR
4	NOT
5	WIRE

$$P1 = \begin{pmatrix} 1 & 1 & 5 \\ 3 & 5 & 0 \\ 0 & 0 & 0 \\ 7 & 3 & 8 \\ 7 & 4 & 0 \\ 0 & 0 & 0 \\ 10 & 2 & 11 \end{pmatrix} \begin{matrix} \rightarrow X1 = A \text{ AND } C \\ \rightarrow X2 = B \\ \rightarrow Y1 = X1 \text{ XOR } X2 \\ \rightarrow Y2 = X1 \\ \rightarrow Z = Y1 \text{ OR } Y2 \end{matrix}$$

Figure 2. The matrix of P1

TABLE II. MATRIX VALUE OF 1ST AND 3RD COLUMN

Matrix value of 1 st column and 3 rd column	INPUT
1	A
2	$\bar{A}$
3	B
4	$\bar{B}$
5	C
6	$\bar{C}$
7	X1
8	X2
9	X3
10	Y1
11	Y2
12	Y3
13	Z

TABLE III. THE TRUE VALUE OF THE EVOLUTIONARY CIRCUIT AND THE DESIRED CIRCUIT

Inputs			Outputs	
A	B	C	Z	Z'
0	0	0	0	0
0	0	1	1	1
0	1	0	1	1
0	1	1	0	0
1	0	0	0	0
1	0	1	1	0
1	1	0	0	1
1	1	1	1	0

Encode the circuit as a matrix in this way. For example, consider the P1 matrix shown in Fig. 2. $P1(1, 1) = 1$ means that the first input of X1 is A. $P1(1, 3) = 5$ means that the second input of X1 is C. $P1(1, 2) = 1$ indicates that the logic gate of X1 is an AND gate. Therefore, $X1 = A \text{ AND } C$. If there is an element in the matrix to be 0, this means that there are no inputs or outputs for this row. For example, $P1(2, 3) = 0$ means there is no input. $P1(3, 2) = 0$ means there is no logic gate.

The particle is then evaluated to see if the circuit represented by the particle satisfies the target truth table. Calculate the energy function value of each particle. The output corresponding to the combined input is encoded as "1" or "-1". If the output Z is 0, the corresponding S_i is "-1", otherwise, it is "1"; if Z' is 1, the corresponding S_j is "1", otherwise, it is "-1". For example, the true value of the evolutionary circuit and the desired circuit is shown in TABLE III, then the energy value of the evolutionary particle P1 is: $E_{(s1)} = -J * \sum S_i * S_j = [(-1) * (-1) + 1*1 + 1*1 + (-1)*(-1) + (-1)*(-1) + 1*(-1) + (-1)*1 + 1*(-1)] = -2J$.

According to the energy value, the circuit with the lowest energy value for each particle is p_best and the circuit with the lowest energy value for all particles is g_best . And then update the particles according to (1), where V_{PN} is initially 0.1, $rand_1$ and $rand_2$ are random numbers between 0.0 and 1.0, $C_1 = C_2 = 2$, W_{min} is 0.4 and W_{max} is 0.9, and the maximum iterations is 3000.

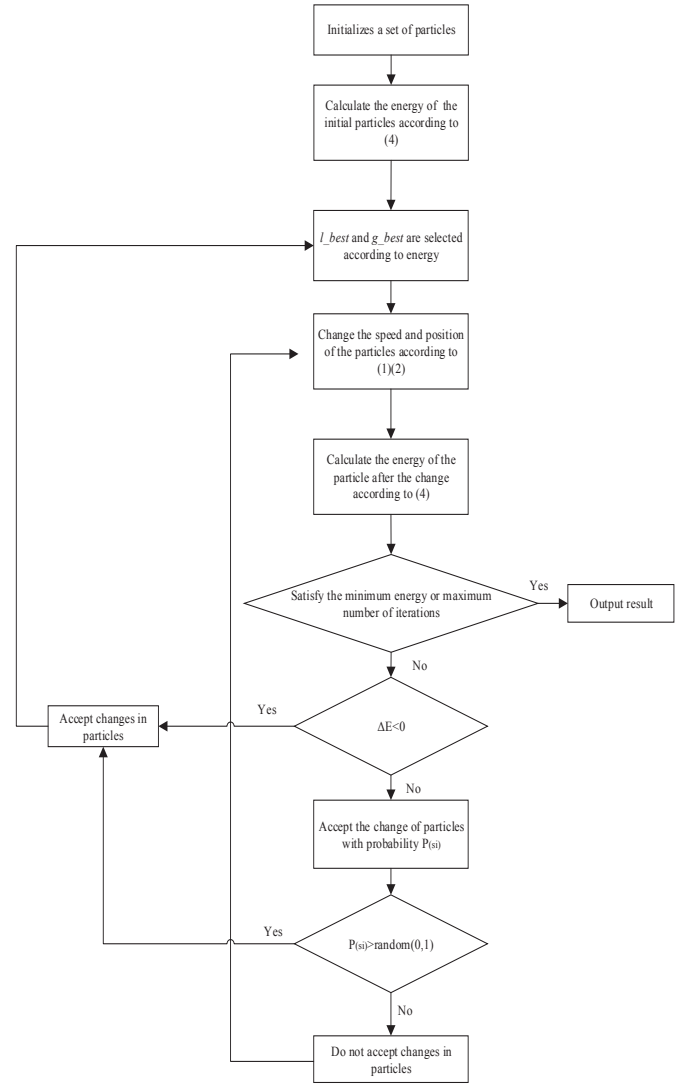


Figure 3. The flow chart of IPSO algorithm

Calculate the energy of each particle after the update, it is determined whether or not the minimum energy is reached. If no, calculate the energy difference value before each particle update, if the energy difference is negative, accept the change of the particles, otherwise, accept the change of the particle with probability $P(si)$. Calculate the probability $P(si)$ from (5). Repeat this process until the termination condition (satisfying the minimum energy value or the maximum number of iterations), and then a combinatorial logic circuit satisfying the target truth table is obtained. The flow chart of the IPSO algorithm is shown in Fig. 3.

IV. EXPERIMENT AND RESULTS ANALYSIS

We experiment with the full adder circuit, it is composed of sum and carry, and its truth table is shown in TABLE IV. The human design circuit performed by the Karnaugh map is shown in Fig. 4.

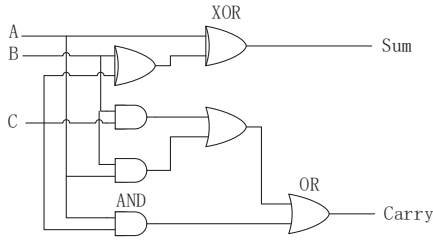


Figure 4. Full Adder circuit designed by using K-map

A. Generate Circuit For Sum

We first initialize five particles P1 to P5 according to the coding rules, as shown below:

$$P1 = \begin{pmatrix} 1 & 1 & 3 \\ 3 & 2 & 5 \\ 0 & 0 & 0 \\ 7 & 3 & 8 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 10 & 4 & 0 \end{pmatrix} \quad P2 = \begin{pmatrix} 1 & 1 & 5 \\ 3 & 1 & 5 \\ 0 & 0 & 0 \\ 7 & 2 & 8 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 10 & 4 & 0 \end{pmatrix} \quad P3 = \begin{pmatrix} 1 & 3 & 3 \\ 1 & 3 & 5 \\ 0 & 0 & 0 \\ 7 & 1 & 8 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 10 & 4 & 0 \end{pmatrix}$$

$$P4 = \begin{pmatrix} 1 & 1 & 5 \\ 1 & 2 & 3 \\ 1 & 3 & 5 \\ 0 & 0 & 0 \\ 8 & 1 & 9 \\ 0 & 0 & 0 \\ 7 & 3 & 11 \end{pmatrix} \quad P5 = \begin{pmatrix} 1 & 2 & 3 \\ 1 & 1 & 5 \\ 3 & 3 & 3 \\ 7 & 3 & 8 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 9 & 3 & 10 \end{pmatrix}$$

The initial energy value of each particle according to (4), as shown in TABLE V. The target truth table is the sum part of TABLE IV; the truth table of the initial particles is shown in TABLE VI.

The IPSO algorithm presented in this paper is used to evolve the initial particles. After each iteration, the energy of each updated particle is calculated and the particle with the lowest energy value is g_best . Judge whether the energy of the particle has reached the minimum value. If not, calculate the energy difference between each particle after the update and before the update. If the energy difference is negative, accept the update change of the particle; otherwise, accept the change of the particle with probability $P_{(si)}$.

TABLE IV. TRUTH TABLE OF FULL ADDER

Inputs			Outputs	
A	B	C	Sum	Carry
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

TABLE V. INITIAL ENERGY OF EACH PARTICLE

	P1	P2	P3	P4	P5
Energy	0	-2	0	-2	0

TABLE VI. THE TRUTH TABLE OF THE INITIAL PARTICLES

Inputs			Outputs				
A	B	C	P1	P2	P3	P4	P5
0	0	0	1	1	1	0	0
0	0	1	0	1	1	0	0
0	1	0	0	1	1	0	1
0	1	1	0	0	0	1	1
1	0	0	1	1	0	1	1
1	0	1	0	0	1	1	0
1	1	0	1	1	1	1	1
1	1	1	1	0	1	1	0

TABLE VII. ENERGY AFTER 11TH ITERATION

	P1	P2	P3	P4	P5
Energy	0	0	-8	0	0

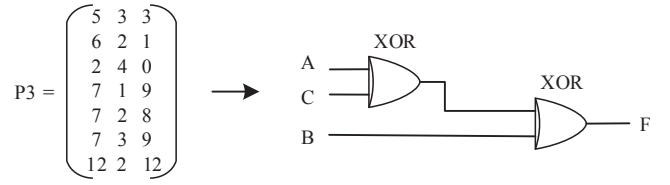


Figure 5. circuit after 11TH iteration

It can be seen from TABLE VII that the minimum energy value is reached after 11 iterations. Particle P3 reaches the minimum energy value. The matrix of P3 after 11 iterations and the evolutionary circuit that meets the target truth table is shown in Fig. 5. And the boolean expression represented by P3 is: $Z=Y3=X1 \text{ XOR } X3=A \text{ XOR } C \text{ XOR } B$.

B. Generate Circuit For Carry

We also use the IPSO algorithm to evolve carry circuit for the five initial matrices. The truth table for the carry circuit is shown in the carry section of TABLE IV.

The IPSO algorithm is used to change the initial particle and make it evolve towards the target truth table. During the evolution process, the change of the particle is determined according to the calculated probability $P_{(si)}$, so as to prevent the particle from falling into the local optimal point for a long time. After several iterations, the energy values of the five particles are finally shown in TABLE VIII.

TABLE VIII. THE FINAL ENERGY VALUE OF EACH PARTICLE

	P1	P2	P3	P4	P5
Energy	-8	-4	-4	0	0

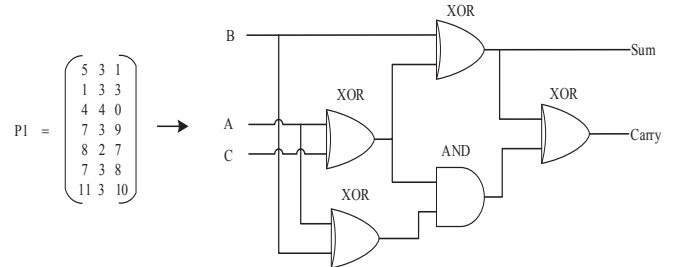


Figure 6. The circuit represented by P1

The matrix of P1 and the evolutionary circuit that meets the target truth table is shown in Fig. 6. Boolean expression represented by P1 is: $Z = Y2 \text{ XOR } Y1 = (X2 \text{ AND } X1) \text{ XOR } (X1 \text{ XOR } X3) = [(A \text{ XOR } C) \text{ AND } (A \text{ XOR } B)] \text{ XOR } [A \text{ XOR } C \text{ XOR } B]$. It can be seen from TABLE VIII that the particle P1 finally reaches the minimum energy.

C. Generate Circuit For Full Adder

Combined with the circuit in Part A and Part B, the complete circuit is shown in Fig. 7.

D. Comparison Of Results

1) Comparison of evolutionary processes:

In the evolution of sum circuit, it is recorded that the fitness/energy values of the g_best particles at each iteration during the evolution of the target circuit for the PSO algorithm and the IPSO algorithm respectively. The evolution process is shown in Fig. 8.

It can be seen in Fig. 8 that the IPSO algorithm jumps out of the local optimum point desired iterations less than the PSO algorithm. It proves that when evolving combinatorial logic circuits, the IPSO algorithm has the advantage of avoiding particles falling into the local optimum compared with the PSO algorithm, thereby reducing the iterations and saving computing resources.

2) Comparison of circuit area:

The circuit designed by the IPSO algorithm proposed in this paper is compared with the circuit designed by human using K_map and the circuit obtained by the PSO algorithm in [10]. The results are shown in TABLE IX , where the number of transistors constituting the AND gate, OR gate, and XOR gate is 6.

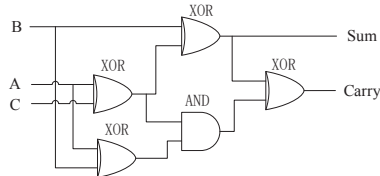


Figure 7. Full adder circuit optimized by IPSO

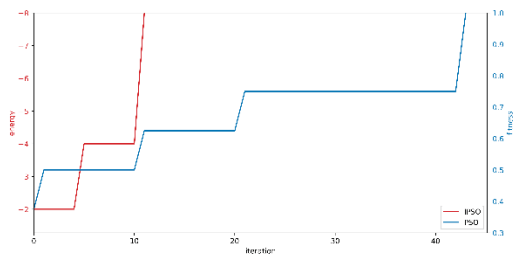


Figure 8. The circuit evolution process of PSO and IPSO

TABLE IX. COMPARISON OF RESULTS

	Human design by using K_map	PSO by [10]	IPSO
Iterations for Sum	--	32	11
Gate count	7	6	5
The number of transistors	42	36	30

V. CONCLUSION

In this work, we propose a hybrid algorithm of Ising model and PSO and apply it to the optimization of combinatorial logic circuits. We can see from the experimental data that the area of our proposed hybrid algorithm is reduced by 28.57% compared with the Karnaugh map human design; compared with the PSO algorithm in [10], the area is reduced by 14.25%. In addition, in the process of sum circuit evolution, the number of iterations required by the IPSO algorithm proposed in this paper is 74.42% less than the PSO algorithm^[10]. This proves that the IPSO algorithm has the advantage of preventing particles from falling into the local optimum during the evolution process, and can reduce the number of iterations to reduce the computational complexity; at the same time, the IPSO algorithm can evolve combinatorial circuits with fewer gates, thereby reducing chip area.

In addition, circuit delay is also an important indicator in digital circuit design. In future work, static timing analysis technology can be combined to take the cell delay into account in circuit evolution, so that a digital circuit that meets timing requirements and has the smallest area can be obtained through intelligent evolutionary algorithms.

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